Residue at $s = \frac{2}{3}$ for a Farey–inverse zeta function

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Abstract

We define a Dirichlet series $Z_f(s)$ built from Farey (Stern–Brocot) intervals and a C^3 function f with $|f''|$ bounded above and below. We show that $Z_f(s)$ admits a meromorphic continuation to a neighborhood of $s = \frac{2}{3}$ with a simple pole there, and we compute its residue as a universal constant times $\int_0^1 |f''|^{2/3}$. All analytic steps are proved in detail. The only arithmetic input is the Weil bound for Kloosterman sums.

1 Farey intervals and the zeta function

Definition 1 (Farey interval)**.** A *Farey interval* is an interval

$$I = [c/d,\, a/b] \subset [0, 1]$$

with $b, d \in \mathbb{N}$, $a, c \in \mathbb{Z}$ satisfying $0 \le c \le d$, $0 \le a \le b$, $\gcd(a, b) = \gcd(c, d) = 1$, and

$$ad - bc = 1.$$

We write its mediant as $m(I) = \frac{a+c}{b+d}$.

Definition 2 (Hata coefficient and its renormalization)**.** Let $f \in C^2([0, 1])$. For $I = [c/d, a/b]$ define

$$c_I(f) := f\left(\frac{a+c}{b+d}\right) - \frac{b}{b+d} f\left(\frac{a}{b}\right) - \frac{d}{b+d} f\left(\frac{c}{d}\right).$$

Define

$$T_I(f) := (b + d)\, c_I(f).$$

Definition 3 (Farey zeta function)**.** For $s \in \mathbb{C}$ with $\Re(s)$ large, define

$$Z_f(s) := \sum_I |T_I(f)|^s,$$

where the sum runs over all Farey intervals $I \subset [0, 1]$.

Remark 1 (On residues)*.* Throughout, $\mathrm{Res}_{s=2/3}$ denotes the residue of a *meromorphic continuation* of the relevant function to a punctured neighborhood of $s = \frac{2}{3}$. Accordingly, whenever we compute a residue we also prove that, after subtracting the explicit polar part, the remainder is holomorphic on some disk $|s - \frac{2}{3}| < \eta$.

Remark 2*.* Removing or adding finitely many intervals (e.g. those adjacent to 0/1 or 1/1) changes $Z_f(s)$ by an entire function. Hence it does not affect the residue at $s = \frac{2}{3}$.

2 Exact mean-value identity

Lemma 1 (Two-point second-order mean value formula)**.** *Let $f \in C^2([0,1])$, let $x \neq y$ in $[0,1]$, and let $\lambda \in (0,1)$. Put $m = \lambda x + (1-\lambda)y$. Then there exists ξ between x and y such that*

$$f(m) - \lambda f(x) - (1-\lambda)f(y) = -\frac{1}{2}\lambda(1-\lambda)f''(\xi)(x-y)^2.$$

Proof. Let $\varphi(t) = f\big(y + t(x-y)\big)$ on $[0,1]$. Then $\varphi \in C^2$ and

$$\varphi(\lambda) - \big(\varphi(0) + \lambda(\varphi(1) - \varphi(0))\big)$$

is the error of linear interpolation at $t = \lambda$. By the standard second-order mean value theorem for the interpolation error, there exists $\eta \in (0,1)$ such that

$$\varphi(\lambda) - \big(\varphi(0) + \lambda(\varphi(1) - \varphi(0))\big) = \frac{1}{2}\varphi''(\eta)(\lambda - 0)(\lambda - 1) = -\frac{1}{2}\lambda(1-\lambda)\varphi''(\eta).$$

Since $\varphi''(t) = f''(y + t(x-y))(x-y)^2$, setting $\xi = y + \eta(x-y)$ yields the claim. □

Corollary 1 (Exact form of $T_I(f)$)**.** *Let $I = [c/d, a/b]$ be a Farey interval and $f \in C^2([0,1])$. Then there exists $\xi_I \in (c/d, a/b)$ such that*

$$T_I(f) = -\frac{f''(\xi_I)}{2\,bd(b+d)}.$$

Proof. Apply Lemma 1 with $x = a/b$, $y = c/d$, and $\lambda = b/(b+d)$, so $m = m(I) = \frac{a+c}{b+d}$. Then

$$(b+d)c_I(f) = -\frac{1}{2}(b+d)\lambda(1-\lambda)f''(\xi_I)(x-y)^2.$$

Now $\lambda(1-\lambda) = \frac{bd}{(b+d)^2}$ and $(x-y)^2 = \frac{(ad-bc)^2}{b^2d^2} = \frac{1}{b^2d^2}$. Substitute and simplify. □

3 Regularity hypotheses and analytic dependence on s

Assume from now on that

$$f \in C^3([0,1]) \quad \text{and} \quad 0 < m \leq |f''(x)| \leq M \text{ for all } x \in [0,1]. \tag{1}$$

For $s \in \mathbb{C}$ define the 1-periodic function

$$B_s(x) := |f''(x)|^s = \exp\big(s \log |f''(x)|\big).$$

Then for each fixed x, $s \mapsto B_s(x)$ is entire. Moreover, for s in any compact set $K \subset \mathbb{C}$ we have uniform bounds

$$\sup_{s\in K} \|B_s\|_\infty \leq M^{\sup_{s\in K} \Re(s)}, \qquad \sup_{s\in K} \|(B_s)'\|_\infty \leq C_K < \infty, \tag{2}$$

since $(B_s)'(x) = s\,|f''(x)|^s \frac{f'''(x)}{f''(x)} \cdot \operatorname{sgn}(f''(x))$ and f'''/f'' is bounded by (1).

4 A model endpoint-sampled series

Define

$$Z_f^{\mathrm{end}}(s) := 2^{-s} \sum_{I=[c/d,a/b]} \frac{|f''(a/b)|^s}{(bd(b+d))^s}.$$

Lemma 2 (Replacing ξ_I by an endpoint does not change the residue)**.** *There exists $\eta_0 > 0$ such that $Z_f(s) - Z_f^{\mathrm{end}}(s)$ is holomorphic on the disk $|s - \frac{2}{3}| < \eta_0$. In particular,* $\mathrm{Res}_{s=2/3} Z_f = \mathrm{Res}_{s=2/3} Z_f^{\mathrm{end}}$.

Proof. By Corollary 1, $Z_f(s) = 2^{-s} \sum_I |f''(\xi_I)|^s/(bd(b+d))^s$. Fix a small disk $D := \{s : |s-2/3| \le \eta\}$ with $0 < \eta < 1/30$. Let $\sigma_- := \min_{s\in D} \Re(s) = 2/3 - \eta$.

For $s \in D$ and any $u, v \in [0, 1]$, by the mean value theorem and (2),

$$\big||f''(u)|^s - |f''(v)|^s\big| \le \sup_{s\in D} \|(B_s)'\|_\infty \, |u - v| \le C_D \, |u - v|.$$

For a Farey interval $I = [c/d, a/b]$, we have $|\xi_I - a/b| \le |a/b - c/d| = 1/(bd)$. Hence for $s \in D$,

$$\left| \frac{|f''(\xi_I)|^s - |f''(a/b)|^s}{(bd(b+d))^s} \right| \le \frac{C_D}{(bd)\,(bd(b+d))^{\sigma_-}} = C_D \, b^{-(\sigma_-+1)} d^{-(\sigma_-+1)} (b+d)^{-\sigma_-}.$$

The right-hand side is independent of s and is summable over all $b, d \ge 1$ because at $\sigma_- = 2/3 - \eta$ the exponent on each of b, d is $\sigma_- + 1 > 1$. Therefore the series defining $Z_f(s) - Z_f^{\mathrm{end}}(s)$ converges uniformly on D by the Weierstrass M-test. Since each term is holomorphic in s, the sum is holomorphic on the interior of D. □

5 Parametrization and the kernel H_s

Lemma 3 (Parametrization by modular inverses)**.** *Farey intervals $I = [c/d, a/b]$ (with $b, d \ge 1$) are in bijection with coprime pairs $(b, d) \in \mathbb{N}^2$ via*

$$a \equiv d^{-1} \pmod{b}, \quad 1 \le a \le b, \qquad c = \frac{ad-1}{b}.$$

Proof. If $ad - bc = 1$, then $ad \equiv 1 \pmod{b}$, so $(b, d) = 1$ and $a \equiv d^{-1} \pmod{b}$. Conversely, for coprime (b, d) choose a with $ad \equiv 1 \pmod{b}$ and define $c = (ad - 1)/b \in \mathbb{Z}$; then $ad - bc = 1$. □

For $s \in \mathbb{C}$ and $u \in (0, 1]$ define

$$H_s(u) := \sum_{k=0}^{\infty} (k+u)^{-s}(k+1+u)^{-s}.$$

Using Lemma 3 and writing $d = kb + r$ with $(r, b) = 1$, one obtains

$$Z_f^{\mathrm{end}}(s) = 2^{-s} \sum_{b\ge1} b^{-3s} \sum_{\substack{1\le r\le b \\ (r,b)=1}} |f''(\overline{r}/b)|^s \, H_s(r/b), \tag{3}$$

where $\overline{r}$ denotes the inverse of r modulo b in $\{1, \ldots, b\}$.

Lemma 4 (Integral of H_s)**.** *For $\frac{1}{2} < \Re(s) < 1$,*

$$\int_0^1 H_s(u)\,du = \int_0^\infty t^{-s}(1+t)^{-s}\,dt = \frac{\Gamma(1-s)\Gamma(2s-1)}{\Gamma(s)}.$$

In particular, $\int_0^1 H_{2/3}(u)\,du = \Gamma(1/3)^2/\Gamma(2/3)$.

Proof. For $\Re(s) > 1/2$, Tonelli implies $\int_0^1 H_s = \sum_{k\geq 0}\int_0^1 (k+u)^{-s}(k+1+u)^{-s}du$. Substitute $t = k+u$ in each term and note that the intervals $[k, k+1]$ partition $[0,\infty)$. This yields $\int_0^\infty t^{-s}(1+t)^{-s}dt$, which equals the Beta integral. □

6 Auxiliary bounds for H_s and reduced-residue sampling

Lemma 5 (Pointwise bound for H_s)**.** *For $1/2 < \sigma := \Re(s) < 1$ and $u \in (0,1]$ one has*

$$|H_s(u)| \leq C_s\, u^{-\sigma},$$

with C_s depending continuously on s on compact subsets of $\{1/2 < \Re(s) < 1\}$.

Proof. Write $H_s(u) = u^{-s}(1+u)^{-s} + \sum_{k\geq 1}(k+u)^{-s}(k+1+u)^{-s}$. The first term has modulus at most $u^{-\sigma}$. For the tail, $\sum_{k\geq 1}(k+u)^{-\sigma}(k+1+u)^{-\sigma} \leq \sum_{k\geq 1} k^{-2\sigma} < \infty$. Since $u^{-\sigma} \geq 1$ on $(0,1]$, the tail is $\ll_s u^{-\sigma}$. □

Lemma 6 (Derivative bound for H_s)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$. Then there exists $C_K > 0$ such that for all $s \in K$ and all $u \in (0,1]$,*

$$\left|\partial_u H_s(u)\right| \leq C_K\, u^{-\Re(s)-1}.$$

Proof. Differentiate termwise (absolute convergence holds uniformly for $u \in [\delta, 1]$, and the $k = 0$ term controls the singularity as $u \downarrow 0$):

$$\partial_u H_s(u) = -s\sum_{k\geq 0}(k+u)^{-s-1}(k+1+u)^{-s} - s\sum_{k\geq 0}(k+u)^{-s}(k+1+u)^{-s-1}.$$

For $u \in (0,1]$ the $k = 0$ contributions satisfy $|(u)^{-s-1}(1+u)^{-s}| + |u^{-s}(1+u)^{-s-1}| \ll_K u^{-\Re(s)-1}$. For the tails $k \geq 1$ we bound each summand by $|s|\,(k+u)^{-\sigma-1}(k+1+u)^{-\sigma} + |s|\,(k+u)^{-\sigma}(k+1+u)^{-\sigma-1} \ll_K k^{-2\sigma-1}$, with $\sigma = \Re(s) > 1/2$, and $\sum_{k\geq 1} k^{-2\sigma-1} < \infty$ uniformly on K. Since $u^{-\sigma-1} \geq 1$ on $(0,1]$, the tail is also $\ll_K u^{-\sigma-1}$. □

Lemma 7 (Reduced residues: H_s averages to its integral)**.** *Fix a compact set $K \Subset \{0 < \Re(s) < 1\}$. Then for $s \in K$ and $b \geq 1$,*

$$\sum_{\substack{1\leq r\leq b\\(r,b)=1}} H_s(r/b) = \varphi(b)\int_0^1 H_s(u)\,du + O_K\big(b^{\Re(s)}\tau(b)\big),$$

where $\tau(b)$ is the divisor function and the implied constant is uniform for $s \in K$.

Proof. By inclusion–exclusion,

$$\sum_{\substack{1\le r\le b\\(r,b)=1}} H_s(r/b) = \sum_{d|b}\mu(d)\sum_{1\le k\le b/d} H_s\left(\frac{k}{b/d}\right).$$

Write $N = b/d$ and denote $S_s(N) := \sum_{k=1}^{N} H_s(k/N)$. We prove

$$S_s(N) = N\int_0^1 H_s(u)\,du + O_K(N^{\Re(s)}), \tag{4}$$

uniformly for $s \in K$.

Decompose $H_s(u) = u^{-s}(1+u)^{-s} + R_s(u)$, where $R_s(u) := \sum_{k\ge1}(k+u)^{-s}(k+1+u)^{-s}$. For $s \in K$, the series for R_s converges uniformly in $u \in [0,1]$ together with its u-derivative, since $|\partial_u(k+u)^{-s}(k+1+u)^{-s}| \ll_K k^{-2\Re(s)-1}$ and $\Re(s) > 0$. Hence $R_s \in C^1([0,1])$ and $\sup_{s\in K}\|R_s'\|_\infty < \infty$. A standard Riemann-sum estimate for C^1 functions yields

$$\sum_{j=1}^{N} R_s(j/N) = N\int_0^1 R_s(u)\,du + O_K(1).$$

It remains to treat the singular part $G_s(u) := u^{-s}(1+u)^{-s}$. Write $G_s(u) = u^{-s} + u^{1-s}Q_s(u)$, where $Q_s(u) := \frac{(1+u)^{-s}-1}{u}$ extends continuously to $u = 0$ with $Q_s(0) = -s$. For $s \in K$, Q_s is C^1 on $[0,1]$ with norms bounded uniformly in s. Therefore

$$\sum_{j=1}^{N}(j/N)^{1-s}Q_s(j/N) = N\int_0^1 u^{1-s}Q_s(u)\,du + O_K(1).$$

So the contribution of $u^{1-s}Q_s(u)$ to (4) is $O_K(1)$.

Finally, for u^{-s} we use a direct Riemann-sum estimate (no zeta-function input). For $s \in K$ (hence $\Re(s) > 0$ and $s \ne 1$), the function $x \mapsto x^{-s}$ is C^1 on $[1,\infty)$ and

$$\sum_{j=1}^{N} j^{-s} = \int_1^N x^{-s}\,dx + O_K(1).$$

Indeed,

$$\sum_{j=1}^{N} j^{-s} - \int_1^N x^{-s}dx = \sum_{j=1}^{N}\int_j^{j+1}\left(j^{-s} - x^{-s}\right)dx + O_K(1),$$

and the sum is bounded in modulus by $\sum_{j\ge1}\int_j^{j+1}\int_j^{j+1}|(x^{-s})'|\,dx\,dt \ll_K \int_1^\infty x^{-\Re(s)-1}dx < \infty$. Multiplying by N^s yields

$$\sum_{j=1}^{N}(j/N)^{-s} = N^s\int_1^N x^{-s}dx + O_K(N^{\Re(s)}).$$

But $N^s\int_1^N x^{-s}dx = \frac{N-N^s}{1-s}$, so

$$\sum_{j=1}^{N}(j/N)^{-s} - N\int_0^1 u^{-s}du = -\frac{N^s}{1-s} + O_K(N^{\Re(s)}) = O_K(N^{\Re(s)}),$$

since $|N^s| = N^{\Re(s)}$. Combining the three pieces proves (4).

Returning to inclusion–exclusion,

$$\sum_{d|b} \mu(d) S_s(b/d) = \left(\sum_{d|b} \mu(d) \frac{b}{d} \right) \int_0^1 H_s + O_K \left(\sum_{d|b} (b/d)^{\Re(s)} \right).$$

The main coefficient equals $\varphi(b)$. The error satisfies $\sum_{d|b} (b/d)^{\Re(s)} \le b^{\Re(s)} \tau(b)$. □

7 Incomplete Kloosterman sums (Weil bound)

For integers n and $b \ge 1$ define the incomplete Kloosterman sum

$$K_b(n; R) := \sum_{\substack{1 \le r \le R \\ (r,b)=1}} e\left(\frac{n\overline{r}}{b} \right), \qquad e(x) := e^{2\pi i x}.$$

Lemma 8 (Completion bound for $K_b(n; R)$)**.** *For any $\varepsilon > 0$,*

$$\max_{1 \le R \le b} |K_b(n; R)| \ll_\varepsilon b^{1/2+\varepsilon},$$

uniformly in n and b. More explicitly one may take the bound $\ll b^{1/2} \tau(b)^2 \log(2b)$.

Proof. Let $w_R(r) = \mathbf{1}_{1 \le r \le R}$ viewed on $\mathbb{Z}/b\mathbb{Z}$, and expand

$$w_R(r) = \frac{1}{b} \sum_{h \bmod b} \widehat{w}_R(h) e\left(\frac{hr}{b} \right), \qquad \widehat{w}_R(h) := \sum_{r=1}^{R} e\left(-\frac{hr}{b} \right).$$

Then

$$K_b(n; R) = \sum_{\substack{r \bmod b \\ (r,b)=1}} w_R(r) e\left(\frac{n\overline{r}}{b} \right) = \frac{1}{b} \sum_{h \bmod b} \widehat{w}_R(h)\, S(n, h; b),$$

where $S(n, h; b) = \sum_{(r,b)=1} e\big((n\overline{r} + hr)/b\big)$ is the classical Kloosterman sum. By the Weil bound (e.g. [1, Ch. 11]), $|S(n, h; b)| \ll \tau(b)(n, h, b)^{1/2} b^{1/2}$. A standard estimate gives $|\widehat{w}_R(h)| \ll \min\{R,\, b/|h|\}$ for representatives $h \in [-b/2, b/2]$. Group h by $d = (h, b)$ to obtain $\sum_{(h,b)=d} |\widehat{w}_R(h)| \ll (b/d) \log(2b)$. Then

$$\sum_{h \bmod b} |\widehat{w}_R(h)|\, |S(n, h; b)| \ll \tau(b) b^{1/2} \log(2b) \sum_{d|b} (n, d)^{1/2} \frac{b}{d}.$$

Since $(n, d)^{1/2} \le d^{1/2}$, the inner sum is $\ll b \sum_{d|b} d^{-1/2} \ll b\tau(b)$. Thus the numerator is $\ll b^{3/2} \tau(b)^2 \log(2b)$ and division by b yields the claim. Finally $\tau(b)^2 \log(2b) \ll_\varepsilon b^\varepsilon$. □

8 A Fejer approximation bound for C^1 functions

Let F_N denote the Fejer kernel on the circle,

$$F_N(t) := \frac{1}{N} \left(\frac{\sin(\pi N t)}{\sin(\pi t)} \right)^2, \qquad t \in [-1/2, 1/2],$$

extended 1-periodically.

Lemma 9 (First moment of the Fejer kernel)**.** *There exists an absolute constant $C > 0$ such that for all $N \geq 2$,*

$$\int_{-1/2}^{1/2} |t|\, F_N(t)\, dt \leq C\, \frac{\log N}{N}.$$

Consequently, if G is 1-periodic and C^1, then its Fejer mean $G * F_N$ satisfies

$$\|G - G * F_N\|_\infty \leq C\, \frac{\log N}{N}\, \|G'\|_\infty.$$

Proof. Since $F_N \geq 0$ and $\int_{-1/2}^{1/2} F_N = 1$,

$$G(x) - (G * F_N)(x) = \int_{-1/2}^{1/2} \big(G(x) - G(x-t)\big) F_N(t)\, dt.$$

By the mean value theorem, $|G(x) - G(x-t)| \leq \|G'\|_\infty |t|$, giving the second inequality once the moment bound is proved.

For the moment bound, split the integral into $|t| \leq 1/N$ and $1/N < |t| \leq 1/2$. For $|t| \leq 1/N$, use $F_N(t) \leq N$ to get $\int_{|t|\leq 1/N} |t| F_N(t) dt \ll N \int_0^{1/N} t\, dt \ll 1/N$. For $1/N < |t| \leq 1/2$, use $|\sin(\pi t)| \gg |t|$ and $|\sin(\pi N t)| \leq 1$ to obtain $F_N(t) \ll 1/(Nt^2)$. Hence $\int_{1/N<|t|\leq 1/2} |t| F_N(t) dt \ll \frac{1}{N} \int_{1/N}^{1/2} \frac{dt}{t} \ll \frac{\log N}{N}$. □

9 Equidistribution estimate for $\Sigma_b(s)$

Define

$$\Sigma_b(s) := \sum_{\substack{1\leq r\leq b\\ (r,b)=1}} H_s(r/b)\, B_s(\overline{r}/b).$$

Proposition 1 (Equidistribution with a power saving)**.** *Fix a compact set $K \Subset \{\frac{1}{2} < \Re(s) < 1\}$ and assume* (1)*. Then for every $\varepsilon > 0$ and $s \in K$ we have*

$$\Sigma_b(s) = \varphi(b) \left(\int_0^1 H_s(u)\, du\right) \left(\int_0^1 B_s(v)\, dv\right) + O_{f,K,\varepsilon}\big(b^{\frac{1+\Re(s)}{2}+\varepsilon}\big),$$

with implied constant uniform for $s \in K$.

Proof. Fix b and write $A_r := H_s(r/b)$. Let $R := \lfloor b^{1/2} \rfloor$.

Step 1: replace B_s by a Fourier truncation. Since $B_s \in C^1$ and (2) holds uniformly for $s \in K$, Lemma 9 implies $\|B_s - (B_s)_N\|_\infty \ll_K \|(B_s)'\|_\infty \log(2N)/N$, uniformly in $s \in K$. Choose $N := \lfloor b^{1/2} \rfloor$. Then

$$\Sigma_b(s) = \sum_{(r,b)=1} A_r\, (B_s)_N(\overline{r}/b) + O_{f,K}\left(\frac{b\, \log(2N)}{N}\right) = \sum_{(r,b)=1} A_r\, (B_s)_N(\overline{r}/b) + O_{f,K}\big(b^{1/2} \log(2b)\big). \tag{$*$}$$

Here we used the crude bound

$$\sum_{\substack{1\leq r\leq b\\ (r,b)=1}} |A_r| \leq \sum_{r=1}^{b} |H_s(r/b)| \ll_K \sum_{r=1}^{b} (r/b)^{-\Re(s)} = b^{\Re(s)} \sum_{r=1}^{b} r^{-\Re(s)} \ll_K b,$$

valid uniformly for $s \in K$ since $\Re(s) > 1/2$.

Step 2: expand $(B_s)_N$ in Fourier modes. Write

$$(B_s)_N(x) = \widehat{B}_s(0) + \sum_{0<|n|<N} \widehat{(B_s)_N}(n) e(nx), \qquad \widehat{B}_s(0) = \int_0^1 B_s.$$

Then

$$\sum_{(r,b)=1} A_r \, (B_s)_N(\overline{r}/b) = \widehat{B}_s(0) \sum_{(r,b)=1} A_r + \sum_{0<|n|<N} \widehat{(B_s)_N}(n) \, S_b(n),$$

where $S_b(n) := \sum_{(r,b)=1} A_r e(n\overline{r}/b)$.

Step 3: the main term. By Lemma 7, uniformly in $s \in K$,

$$\sum_{(r,b)=1} A_r = \varphi(b) \int_0^1 H_s(u) \, du + O_K\big(b^{\Re(s)} \tau(b)\big).$$

Multiplying by $\widehat{B}_s(0) = \int_0^1 B_s$ gives the desired main term plus an error $O_{f,K}(b^{\Re(s)}\tau(b))$, which is dominated by the final error term since $\Re(s) < 1$.

Step 4: bound the oscillatory sums $S_b(n)$. Split $S_b(n) = S_b^{\le R}(n) + S_b^{>R}(n)$ with the cutoff at R. Using Lemma 5,

$$|S_b^{\le R}(n)| \le \sum_{r \le R} |A_r| \ll_K \sum_{r \le R} (r/b)^{-\Re(s)} \ll_K b^{\Re(s)} R^{1-\Re(s)} \ll_K b^{\frac{1+\Re(s)}{2}}.$$

For the tail we use summation by parts. Let

$$a_r := \mathbf{1}_{(r,b)=1} \, e\left(\frac{n\overline{r}}{b}\right), \qquad A(t) := \sum_{1 \le r \le t} a_r.$$

Then $S_b^{>R}(n) = \sum_{r=R+1}^{b} H_s(r/b) \, a_r$. By Abel summation,

$$S_b^{>R}(n) = H_s(1) \, A(b) - H_s(R/b) \, A(R) - \sum_{t=R}^{b-1} A(t) \Big(H_s\Big(\frac{t+1}{b}\Big) - H_s\Big(\frac{t}{b}\Big) \Big).$$

By Lemma 8 we have $\max_{1 \le t \le b} |A(t)| \ll_\varepsilon b^{1/2+\varepsilon}$. Moreover, Lemma 5 gives $|H_s(R/b)| \ll_K (R/b)^{-\Re(s)}$. For the difference, the mean value theorem together with Lemma 6 yields, uniformly for $s \in K$,

$$\left| H_s\Big(\frac{t+1}{b}\Big) - H_s\Big(\frac{t}{b}\Big) \right| \le \frac{1}{b} \sup_{u \in [t/b, (t+1)/b]} \big|\partial_u H_s(u)\big| \ll_K \frac{1}{b} \left(\frac{t}{b}\right)^{-\Re(s)-1} = O_K\Big(b^{\Re(s)} t^{-\Re(s)-1}\Big).$$

Therefore

$$\sum_{t=R}^{b-1} \left| H_s\Big(\frac{t+1}{b}\Big) - H_s\Big(\frac{t}{b}\Big) \right| \ll_K b^{\Re(s)} \sum_{t \ge R} t^{-\Re(s)-1} \ll_K b^{\Re(s)} R^{-\Re(s)}.$$

Combining these bounds (and using $|H_s(1)| \ll_K 1$) we obtain

$$|S_b^{>R}(n)| \ll_{K,\varepsilon} b^{1/2+\varepsilon} \, b^{\Re(s)} R^{-\Re(s)} \ll_{K,\varepsilon} b^{\frac{1+\Re(s)}{2}+\varepsilon}.$$

Together with the bound for $S_b^{\le R}(n)$ this gives, for $0 < |n| < N$,

$$S_b(n) \ll_{K,\varepsilon} b^{\frac{1+\Re(s)}{2}+\varepsilon}. \tag{5}$$

Step 5: sum Fourier modes. Since $B_s \in C^1$ uniformly in $s \in K$, we have $|\widehat{B}_s(n)| \ll_K \|(B_s)'\|_{L^1}/|n| \ll_K 1/|n|$. Also $|\widehat{(B_s)_N}(n)| \le |\widehat{B}_s(n)|$. Hence

$$\sum_{0<|n|<N} |\widehat{(B_s)_N}(n)| \ll_K \sum_{1\le n<N} \frac{1}{n} \ll \log(2N) \ll_\varepsilon b^\varepsilon.$$

Combining with (5) yields

$$\sum_{0<|n|<N} \widehat{(B_s)_N}(n)\, S_b(n) = O_{f,K,\varepsilon}\big(b^{\frac{1+\Re(s)}{2}+\varepsilon}\big).$$

Finally, the truncation error $O_{f,K}(b^{1/2})$ in $(*)$ is dominated by the same bound. □

10 Meromorphic continuation and residue at $s = \frac{2}{3}$

Theorem 1 (Residue formula)**.** *Assume* (1)*. Then $Z_f(s)$ admits a meromorphic continuation to a disk $|s - \frac{2}{3}| < \eta$ (for some $\eta > 0$) with a simple pole at $s = \frac{2}{3}$. Moreover,*

$$\operatorname{Res}_{s=2/3} Z_f(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{2/3}\pi^3} \int_0^1 |f''(v)|^{2/3}\, dv.$$

Proof. By Lemma 2, it suffices to compute the residue of $Z_f^{\mathrm{end}}(s)$. Fix a disk $D := \{s : |s-2/3| \le \eta\}$ with $0 < \eta < 1/30$ and set $\sigma_- := 2/3 - \eta$, $\sigma_+ := 2/3 + \eta$.

From (3) and Proposition 1 (applied on a compact set containing D but lying in $\{1/2 < \Re(s) < 1\}$), we get for $s \in D$:

$$Z_f^{\mathrm{end}}(s) = 2^{-s}\Big(\int_0^1 H_s\Big)\Big(\int_0^1 B_s\Big)\sum_{b\ge1} \frac{\varphi(b)}{b^{3s}} + E(s),$$

where

$$E(s) = 2^{-s}\sum_{b\ge1} b^{-3s}\, R_b(s), \qquad |R_b(s)| \ll_{f,D,\varepsilon} b^{\frac{1+\Re(s)}{2}+\varepsilon}.$$

In particular, for $s \in D$,

$$|b^{-3s}R_b(s)| \ll_{f,D,\varepsilon} b^{-3\sigma_- + \frac{1+\sigma_+}{2}+\varepsilon} = b^{-\frac{7}{6}+\frac{7\eta}{2}+\varepsilon}.$$

Choose $\varepsilon > 0$ so small that $-\frac{7}{6} + \frac{7\eta}{2} + \varepsilon < -1$ (possible since $\eta < 1/30$). Then $\sum_{b\ge1} b^{-7/6+7\eta/2+\varepsilon}$ converges. Hence the series defining $E(s)$ converges uniformly on D by the Weierstrass M-test. Since each term is holomorphic in s, $E(s)$ is holomorphic on the interior of D.

The factor $A(s) := 2^{-s}\big(\int_0^1 H_s\big)\big(\int_0^1 B_s\big)$ is holomorphic on $\{1/2 < \Re(s) < 1\}$ by Lemma 4 and dominated convergence. The Dirichlet series identity $\sum_{n\ge1}\varphi(n)n^{-w} = \zeta(w-1)/\zeta(w)$ valid for $\Re(w) > 2$ gives a meromorphic continuation of $\sum_{b\ge1}\varphi(b)b^{-3s}$ near $s = 2/3$, with a simple pole at $3s = 2$ and residue $1/(3\zeta(2))$. Therefore $Z_f^{\mathrm{end}}(s)$ (hence $Z_f(s)$) is meromorphic on a neighborhood of $2/3$ with

$$\operatorname{Res}_{s=2/3} Z_f(s) = A(2/3)\cdot\frac{1}{3\zeta(2)}.$$

Now $A(2/3) = 2^{-2/3}\left(\int_0^1 H_{2/3}\right)\left(\int_0^1 |f''(v)|^{2/3}dv\right)$. By Lemma 4, $\int_0^1 H_{2/3} = \Gamma(1/3)^2/\Gamma(2/3)$. Using $\zeta(2) = \pi^2/6$ and $\Gamma(1/3)\Gamma(2/3) = 2\pi/\sqrt{3}$, the constant simplifies to $\sqrt{3}\,\Gamma(1/3)^3/(2^{2/3}\pi^3)$. □

11 Legendre transform and equi-affine length

Corollary 2. *Let $g \in C^3([0,1])$ be strictly concave with $0 < c_1 \le -g'' \le c_2$, and let $f = g^*$ in the tangent-line convention. Then*

$$\int_0^1 |f''(\alpha)|^{2/3}\, d\alpha = \int_0^1 |g''(x)|^{1/3}\, dx.$$

Proof. Under the change of variables $\alpha = g'(x)$ one has $d\alpha = g''(x)\, dx$ and $f''(\alpha) = 1/g''(x)$. Hence $\int |f''(\alpha)|^{2/3}\, d\alpha = \int |1/g''(x)|^{2/3} |g''(x)|\, dx = \int |g''(x)|^{1/3} dx$. □

Example: $g(x) = \frac{2x-1+\sqrt{5-4x}}{2}$

For $g(x) = \frac{2x-1+\sqrt{5-4x}}{2}$ on $[0,1]$, $g''(x) = -\frac{2}{(5-4x)^{3/2}}$. Thus

$$\int_0^1 |g''(x)|^{1/3} dx = 2^{1/3} \int_0^1 (5-4x)^{-1/2} dx = 2^{-2/3}(\sqrt{5}-1).$$

Therefore, with $f = g^*$,

$$\mathrm{Res}_{s=2/3}\, Z_f(s) = \frac{\sqrt{3}\,\Gamma(1/3)^3}{2^{4/3}\pi^3}(\sqrt{5}-1).$$

References

[1] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, American Mathematical Society Colloquium Publications, Vol. 53, 2004.